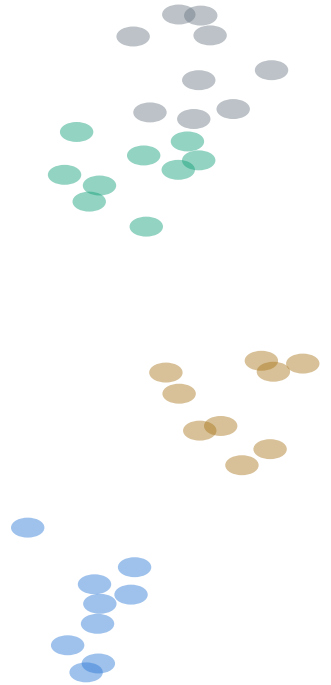
Physics-Defined Tokenization for AI-Ready HEP Datasets

Six experimental campaigns: the particle-library objective, discrete and continuous bottlenecks, and alternate tokenization schemes.



heptokens working group

Data: CMS Aspen Open Jets RunG (1.96 M jets) · 620 k-jet 5-class benchmark

August 2026

Outline

1 Introduction — the AI-ready-dataset mission and the tokenizer question

2 Data, objectives and evaluation methodology — what we measure, and why

3 Campaigns 1–5 — generic SSL, the particle library, hybrid objectives, compression, VAE vs VQ

4 Campaign 6 — alternate tokenization schemes: hypotheses, results, systematic effects

5 Summary — revised conclusions, open problems, outlook

Motivation: the token dataset is the product

- Experiments reconstruct particle-flow candidates that external groups cannot access; those groups want to train foundation models on the events. We publish a per-constituent token as the shareable artifact — versioned by dimension, preprocessing hash and checkpoint.
- Release requirements: information sufficiency for unknown future tasks; detector abstraction; an auditable statement of what the token forgets; compression at the 178 M-jet scale.
- Baseline tokenizer (pre-study): a residual VQ-VAE emitting integer indices. Known concerns: codebook collapse, an implicit metric defined by the reconstruction objective, and lossiness by construction.

RESEARCH QUESTION

Which objective and which bottleneck should define the released per-particle token —
and can any single token be simultaneously compact, information-faithful, and criterion-organized?

Datasets and fairness protocol

Datasets

- jets5c — 620 k jets, 5 balanced classes, ≤150 constituents × (pT, Δη, Δφ). Labelled: used for consumer-view downstream probes.
- Aspen Open Jets RunG — 1.96 M real CMS 2016 jets; constituents (pT, η, φ, PID, charge), PDG ∈ {±211, ±13, ±11, 22, 130}. Unlabelled: all criterion evaluations.

Protocol (held fixed across every comparison)

- Matched budgets: identical encoders, inputs, splits (70/15/15, seed 42), schedules (15–20 epochs); only the objective/bottleneck under test differs.
- Consumer view: downstream probes see only the exported token files — never the constituents.
- Untrained nulls are mandatory: PID-one-hot ⊕ scaled kinematics (192 b); from C6, a criterion-aligned feature grid (~23–26 b). Learning must beat the null to claim rent.
- Seeds: 3 seeds wherever a comparison is promoted to a claim; single-seed numbers are flagged.

hardware: single A100 MIG 2g.20gb slice · torch 2.5.1 · full config in benchmarks/

Objective definitions

PARTICLE LIBRARY (method of record, C2)

$$\text{Positives:}\quad \text{same PID} \;\wedge\; \text{same charge} \;\wedge\; \Delta p_T/p_T \le 0.10 \;\wedge\; |\Delta\eta| \le 0.1 \;\wedge\; |\Delta\varphi| \le 0.1$$

- Context-free per-particle MLP Φ (PID/charge embedded); the same particle maps to the same token in any event.
- Graded Gaussian kernel soft-target contrastive loss: hard gates on PID/charge, smooth grading on the kinematic windows. No augmentations, no views — the metric is stated in physics variables.

VQ-VAE BASELINE (matched, C2b)

- Same encoder → residual VQ ($n_q \times K$ codebook, straight-through + EMA) → decoder reconstructing the full particle (L1 kinematics + CE PID + CE charge). Rate = $n_q \cdot \log_2 K$ bits/particle.
- Deliberately not straw-manned: PID and charge are forced through the bottleneck.

Design rule (project convention): before training, state what each objective makes the embedding forget; generic SSL recipes appear only as ablations.

Primary metric: availability-aware criterion recall@8

$$\text{recall@8} = \text{mean over anchors } a \text{ with } \geq 1 \text{ partner of } \frac{|\text{top-8 neighbours}(a) \cap \text{criterion-partners}(a)|}{\min(8, \#\text{partners}(a))}$$

cosine neighbours in a 30 k-constituent pool · criterion partners computed exactly from raw features

Why not precision@k?

- Criterion density in the pool is $\sim 3 \cdot 10^{-5}$: a typical anchor has far fewer than k true partners available, so precision@k is bounded by availability, not by embedding quality — it flattens all methods to the same small number.
- Recall against the availability ceiling (divide by min(k, #partners)) is 1.0 for a perfect embedding at any density, and it is what the release promises: partners that exist are found.
- recall@8 is the library objective's native metric — it is flagged as such in every table, and reconstruction-flavoured metrics are always reported beside it.

a tighter variant ($\Delta pT \leq 5\%$) is reported as recall@8-dpt5; both are computed by benchmarks/eval_library.py

The metric suite: each entry guards against a known failure

METRIC	WHAT IT MEASURES	FAILURE IT GUARDS AGAINST
criterion recall@8	criterion geometry (native)	precision@k availability ceiling (~$3 \cdot 10^{-5}$ density)
kNN PID purity	species organization (neutral)	alone it misleads: 0.996 purity with 0.383 recall (C5)
within-PID pT R^2	kinematic information retention	catches PID-shortcut tokens; release needs observables
linear PID/charge probes	consumer linear decodability	library structure is metric, not linear (probe ≈ 0.37)
effective rank	dimensional collapse	VICReg/VQ collapse modes (C1: quantizer-0 <10 % codes)
bits / particle	release rate	$n_q \cdot \log_2 K$ · fp16 dim·16 · KL/ln2 (caveat: slide 18)
downstream probe (jets5c)	consumer value of the export	tokens are the product; probes never see constituents

No single metric is trusted alone: every campaign table reports the full suite, with the objective-native column flagged.

Campaign 1 — generic SSL vs VQ-VAE (jets5c, consumer view)

Setup: identical 4-layer transformer encoders; smearing-VICReg contrastive (dim 16) vs residual VQ (512×3, dim 16); probes see only the exported tokens.

INTERFACE	PROBE ACC	MACRO AUC	KNN CLASS PURITY
raw features (ceiling)	0.743	0.931	0.249
contrastive tokens	0.701	0.917	**0.486**
VQ tokens (baseline)	**0.748**	**0.934**	0.243

- The augmentation list silently defines the metric: smearing pT traded away pT sensitivity (distance–ΔpT correlation 0.31 vs 0.61) and cost ≈1.5 AUC downstream.
- VQ pathology observed live: quantizer 0 collapsed to <10 % codebook utilization.
- Anti-collapse machinery works (effective rank 8.3/16, all dims active) — collapse is not the argument against continuous tokens; the implicit metric is.

620 k jets · 20 epochs · matched budgets · probes: 2-layer transformer on 250 k jets

Campaign 2 — the particle library on real CMS data (AOJ)

Setup: the explicit physics criterion replaces the augmentation-implied metric; context-free MLP; no labels, no views. Evaluation: criterion recall@8 in a 30 k pool.

PROVIDER (AOJ)	RECALL@8	KNN PURITY	PT R^2
library (graded kernel)	**0.9992**	0.910	0.18
library (hard windows)	0.9935	0.664	0.30
null: PID-one-hot ⊕ kinematics	0.9833	0.993	**0.95**
raw kinematics only	0.300	0.290	0.95
v1 jet-context transformer	**0.052**	0.284	0.84

- Criterion achieved: 20× fewer retrieval misses than the untrained null — the learning pays rent on the criterion itself. Graded kernel beats hard windows on every axis (no window-edge cliff).
- Context-free is decisive: jet-context encoding destroys the library property (0.052).
- Known deficiency at this stage: within-species pT recovery R^2 0.18 → motivates Campaign 3.

AOJ RunG · 1.96 M jets · 15 epochs · seed 42

Campaign 2b — matched library vs VQ: complementarity

Setup: same data, inputs, encoder, schedule; only the objective differs. 3 seeds each; the VQ arm reconstructs the full particle so identity must pass the bottleneck.

INTERFACE	RECALL@8	PURITY	PID PROBE	PT R^2	BITS
library	**0.9991±.0002**	0.905	0.366	0.169	256
VQ, matched (RVQ 3×512)	0.9271±.0062	**0.996**	**1.000**	**0.971**	**27**
library → post-hoc RVQ	0.832	0.863	0.380	0.174	27
null (untrained)	0.9833	0.993	1.000	0.949	–

- Complementary, not competing: the library owns criterion geometry (~80× fewer misses); the VQ owns information sufficiency (PID/charge linearly decodable, pT R^2 0.97).
- Library species structure is metric, not linear: kNN purity 0.91 but linear PID probe ≈ majority class.
- No downstream cost: library tokens probe at the raw-feature ceiling on jets5c (0.744 acc / 0.931 AUC).

AOJ · 3 seeds · matched ParticleEncoder MLP 128-128-64, dim 16

Campaign 3 — hybrid objective and token dimensionality

Question: can a reconstruction term (weight w) restore kinematic information without breaking the criterion — and how small can the token be?

RUN (AOJ)	RECALL@8	PT R^2	KNN PURITY
kernel only, dim 16	**0.9992**	0.175	0.906
hybrid w=1.0	0.9984	0.867	0.915
hybrid w=3.0	0.9955	**0.908**	0.935
hybrid w=2.0, dim 8	0.9982	0.613	**0.987**
hybrid w=2.0, dim 4	0.9801	0.799	0.987

- w is a smooth, controllable frontier between organization and information — but no single weight reaches the target corner (recall ≥ 0.999 ∧ R^2 ≥ 0.9): best point misses recall by ~0.004.

- The token halves for free: 16→8 dims raises recall and purity; dim 4 is the floor.

AOJ · 15 epochs · seed 42 · target corner quantified by C2b

Campaign 4 — compression: shape the codebook, or shrink the token?

Question: the continuous token is anti-compressive (96–256 b vs ~50 b raw). Does criterion-shaping a VQ codebook give a compact token that keeps the criterion?

ARTIFACT	BITS	RECALL@8	PT R^2
library dim 6 (continuous)	**96**	**0.9996**	**0.743**
library dim 4 (continuous)	64	0.9807	0.684
plain VQ (RVQ 3×512)	27	0.927±.006	0.971
contrastive-VQ, kernel→z_q	27	**0.708**	0.719
contrastive-VQ, kernel→z_e	27	**0.772**	0.295
VQ single 8192 codebook	13	**0.028**	0.821

- Headline negative: the kernel makes the VQ worse than plain reconstruction at equal rate, on both placements — the kernel spreads representation to effective rank 15+, which quantization cannot tile.
- Compression works on the continuous side: dim-6 (96 b) sharpens the criterion and raises pT R^2.
- Granularity beats bit-count: the single 8192 codebook collapses to 1.7 % utilization.

AOJ · 15 epochs · seed 42 (plain VQ: 3 seeds) · 8192 arm at batch 256 (memory) — flagged as a confound in C6 review

Campaign 5 — a 2×2×2 factorial: bottleneck × objective × context

Factors: bottleneck {continuous VAE · RVQ 3×1024 = 30 b} × objective {reconstruction · + physics kernel on the latent} × encoder {context-free MLP · attention}. Everything else matched.

ARTIFACT (AOJ)	BITS	RECALL@8	PT R^2
VAE dim 4, recon only	2.6 (KL)	**0.383**	0.836
VAE dim 4 + kernel	1.4 (KL)	**0.916**	0.665
plain VQ 3×1024	30	**0.945**	**0.970**
contrastive-VQ (kernel→z_e)	30	**0.827**	0.331
VAE recon + attention	2.6 (KL)	**0.051**	0.479
plain VQ + attention	30	0.967	0.967

- Placement is decisive: the same kernel lifts the continuous latent (0.383→0.916) and hurts the codebook (0.945→0.827) — the clean-gradient continuous quadrant is where physics shaping works.

- Context is asymmetric: it helps the 30-bit VQ slightly but destroys the low-rate token (0.383→0.051).

AOJ · 20 epochs · seed 42 · VAE rate = KL/ln2 (revisited on slide 18)

Interim summary after five campaigns

i **Explicit physics criterion ≫ augmentation-implied metric**
0.9992 vs 0.052; the augmentation list is a hidden metric choice.

ii **Criterion and information are separable currencies**
library vs VQ complementarity; w-dial frontier; corner unreached.

iii **The contrastive term needs a smooth continuous latent**
works on μ, hurts through STE+EMA codebooks (twice, both placements).

iv **Compress the continuous token; do not quantize the criterion**
dim-6 at 96 b dominates the VQ compression curve.

v **Context-free encoding preserves the library property**
required for the cross-experiment goal; context helps only at high rate.

OPEN PROBLEM

No trained token is simultaneously sub-raw (≲50 b), faithful ($R^2 \geq 0.9$), and criterion-perfect (recall ≥ 0.999).

Alternate tokenization schemes: what each one probes

A full-repo audit + 2023–26 literature survey (discrete / continuous / HEP-specific), then an adversarial pass over every recorded conclusion. Each scheme isolates one mechanism:

SCHEME	MECHANISM ISOLATED	STATUS
β-sweep (rate-matched VAE)	is the C5 VAE-vs-VQ verdict a rate-mismatch artifact? traces the R–D–criterion surface	**→ 6a**
physics grid	does any learned discrete token beat untrained, criterion-aligned binning of raw features?	**→ 6b**
FSQ (codebook-free)	was the criterion-shaped-VQ failure codebook/EMA dynamics — or quantization itself?	**→ 6c**
rotation-trick · SimVQ	gradient geometry through the quantizer; dead-code dynamics behind q0 collapse	queued
Gumbel / dVAE	stochastic relaxation as the discrete analogue of the clean-gradient placement result	queued
REPA distillation	frozen-target regression vs live co-training: is the C3 corner an optimization artifact?	queued
symbols ⊕ latent	exact categorical channel: restores linear decodability by construction	queued
flow tokens	bijective map ⇒ lossless; how much criterion fits in the leading dimensions?	queued
masked-continuous FM probe	which export does a foundation-model consumer actually prefer?	**→ C7**

Campaign 6 design: three arms, pre-registered hypotheses

6a **Rate-matched contrastive-VAE frontier**

β × dim × free-bits sweep, kernel on the posterior mean μ; 11 arms.

H1: at VQ-level rate the VAE matches VQ on information (bar: recall ≥ 0.945 ∧ R^2 ≥ 0.97 at ≤ 30 b).

6b **Criterion-aligned physics grid (untrained)**

log-pT bins of ln(1.1), η/φ bins of 0.1, exact PID/charge; a half-bin-offset second grid repairs boundary pairs. Evaluation-only.

H2: trained discrete tokens outperform criterion-aligned binning (any learned token below the grid is retired).

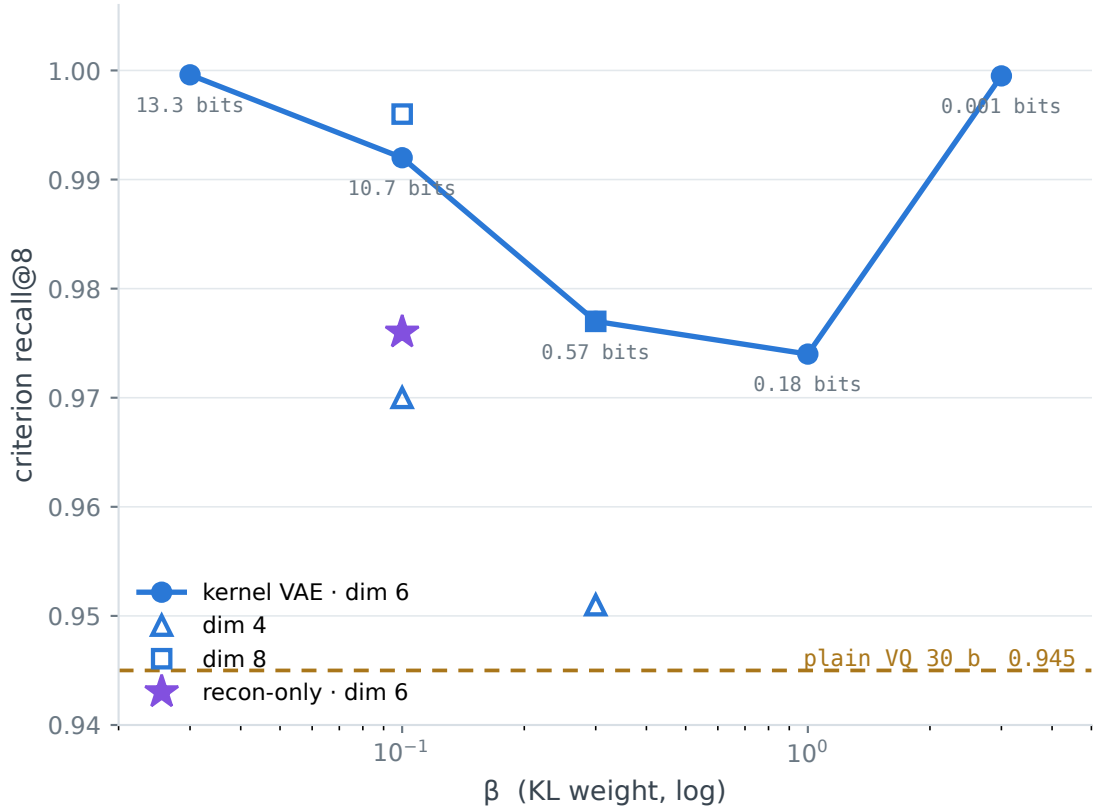
6c **FSQ-bottleneck library tokenizer**

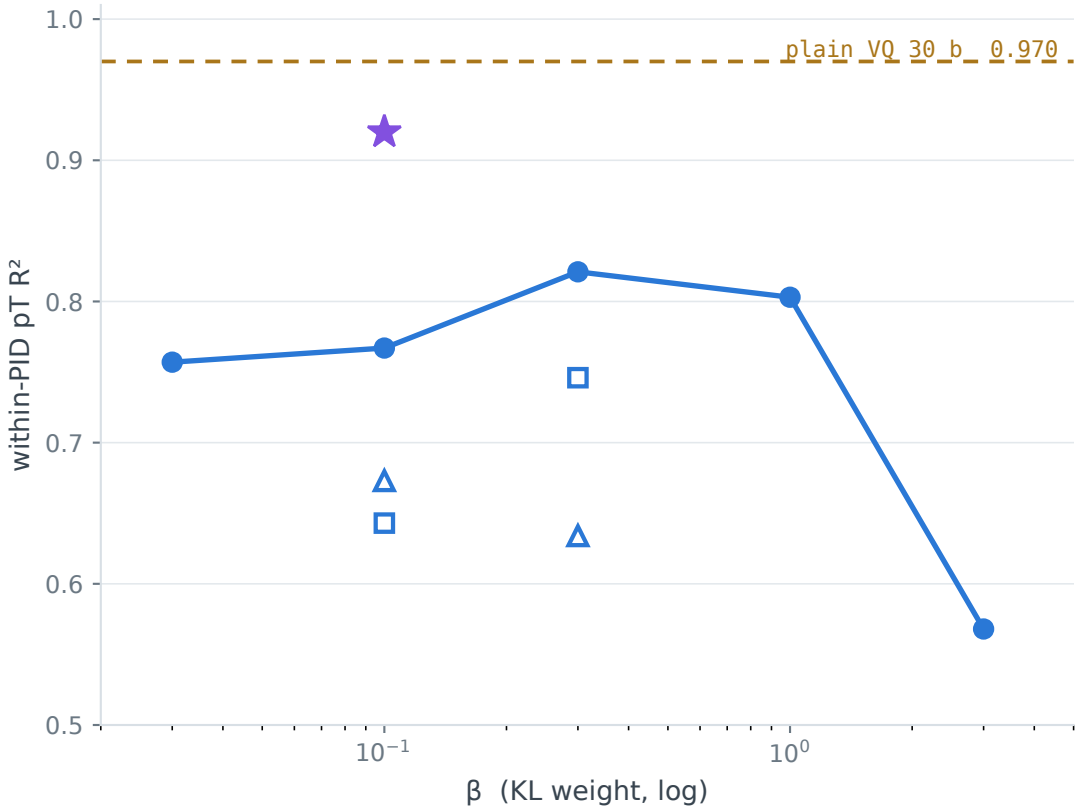
per-dimension rounding of a bounded latent; zero learned parameters, exact designer-chosen bits; kernel on pre-rounding z_e (z_q and reconstruction ablations); 8 arms.

H3: the C4/C5 negative was codebook/EMA dynamics — removing them rescues the criterion-shaped discrete token.

23 training runs · AOJ · context-free MLP · 20 epochs · headline arms at 3 seeds · outcomes pre-registered in docs/research/campaign6_alternate_schemes.md

6a — recall and pT R² along the β dial (H1: rejected)





Recall reaches 0.9984 ± 0.0015 at β = 0.03 (3 seeds) — parity with the 96-bit deterministic dim-6 library — but R² peaks at ≈ 0.82: the retirement criterion (R² ≥ 0.97) is not met at any β. Complementarity survives rate-matching.

Systematic effect: ELBO rate vs information in a deterministic release

OBSERVATION (β = 3, dim 6)

measured KL rate ≈ 0.001 bits/particle — yet criterion recall@8 = 0.9995

- Mechanism: the released token is the posterior mean μ, a deterministic function of the input. At high β the posterior collapses onto the prior (KL → 0) while μ retains a tiny-norm but structured layout — and cosine retrieval is scale-invariant, so the criterion geometry survives at 'zero' rate.
- The KL rate correctly prices transmitting posterior samples (bits-back), not the deterministic μ we actually export. It is therefore not a valid release-rate claim for this token family.
- Consequence for prior results: Campaign 5's '~1.4-bit' contrastive VAE was this effect in milder form. Until an operational entropy codec exists, continuous rates are quoted as fp16 dim×16, and KL-bits are reported as a diagnostic only. free-bits (0.05/dim) does not repair the accounting (0.41 b, 0.974).

all rate columns in the campaign-6 tables and figures follow this convention

Revisiting C5: 'reconstruction buys species, not the criterion'

ARM	BITS	RECALL@8	PT R^2	PID PROBE
recon-only VAE, β=1 (C5)	2.6 (KL)	**0.383**	0.836	1.00
recon-only VAE, β=0.1 (C6)	11.0 (KL)	**0.976±.004**	**0.920±.011**	1.00
null (untrained)	—	0.9833	0.949	1.00
library / kernel VAE (β=.03)	96 (fp16)	**0.9984±.0015**	0.758	0.59

- The C5 conclusion was a rate artifact, not a property of the objective: at β = 0.1 the recon-only VAE recovers 0.976 recall with 0.920 R^2 and perfect linear probes (3 seeds).
- It still does not exceed the untrained null on either axis — the physics kernel remains the only objective that has (0.9984 vs 0.9833).
- Methodological lesson: single-β conclusions about objectives conflate the objective with its rate.

AOJ · 20 epochs · 3 seeds on both revised arms

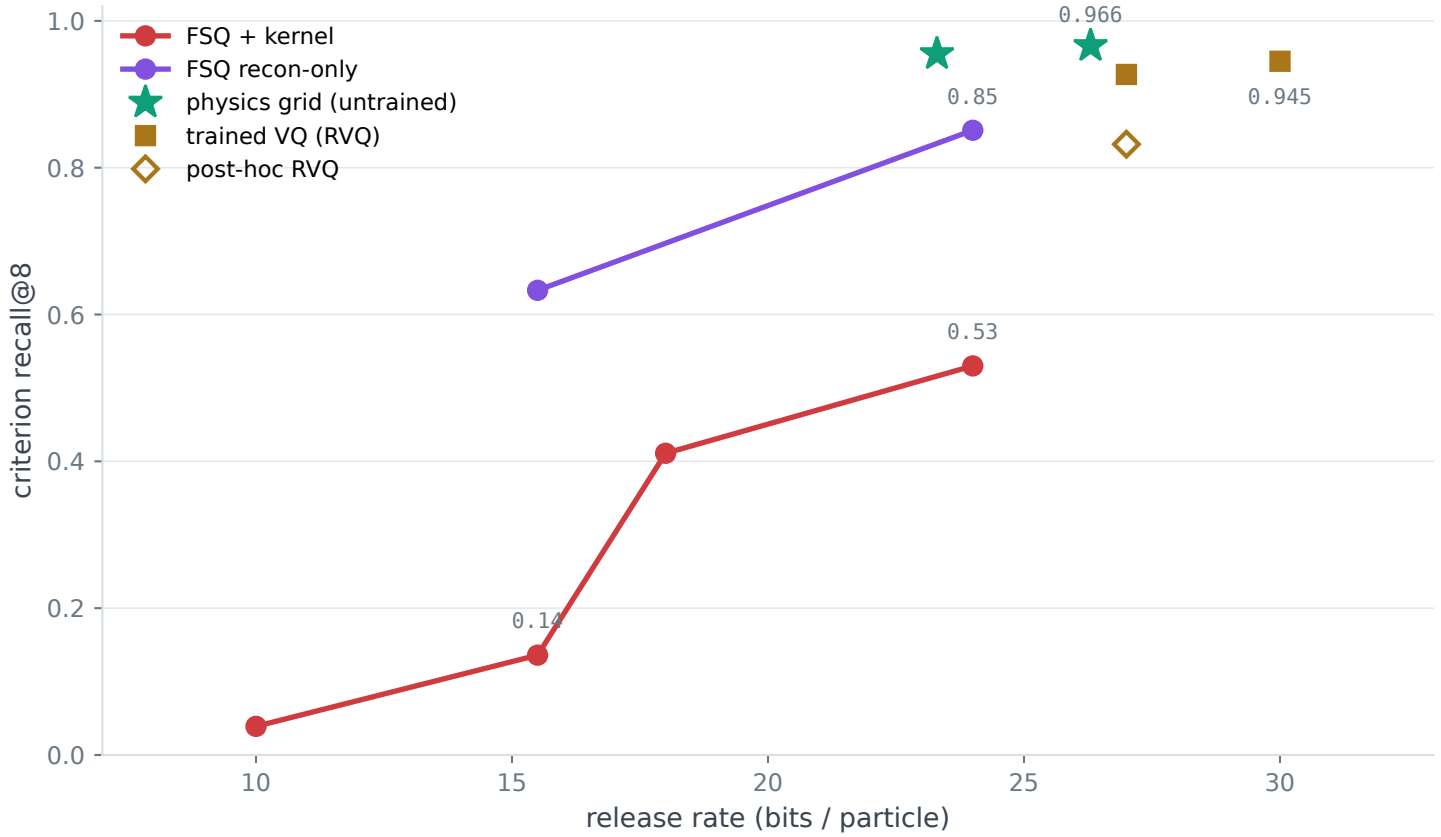
6b — the untrained physics grid vs learned discrete tokens (H2: rejected)

DISCRETE INTERFACE	BITS	RECALL@8	PT R^2	PID PROBE
physics grid, double	**26.3**	**0.9655**	0.949	1.00
physics grid, single	23.3	0.9553	0.948	1.00
plain VQ 3×1024 (trained)	30	0.945	**0.970**	1.00
plain VQ 3×512 (trained)	27	0.927±.006	0.971	1.00
library → post-hoc RVQ	27	0.832	0.174	0.38
FSQ recon-only (trained)	24	0.851	0.840	0.81

- The zero-parameter grid exceeds every learned discrete token on recall at equal or fewer bits; the trained VQ's surviving advantage is +0.02 pT R^2 and +0.017 purity.
- At the 23–30-bit tier, training a discrete token currently buys almost nothing over criterion-aligned binning — and the grid is computable on any detector, a natural cross-experiment release tier.
- H2 fails below the pre-registered auto-retirement threshold (0.99) but the practical ordering stands.

grid = ln(1.1) log-pT × 0.1 η × 0.1 φ bins ⊕ exact PID/charge; double grid adds a half-bin-offset copy (~+3 b)

6c — FSQ: criterion-shaping fails at every rate (H3: rejected)

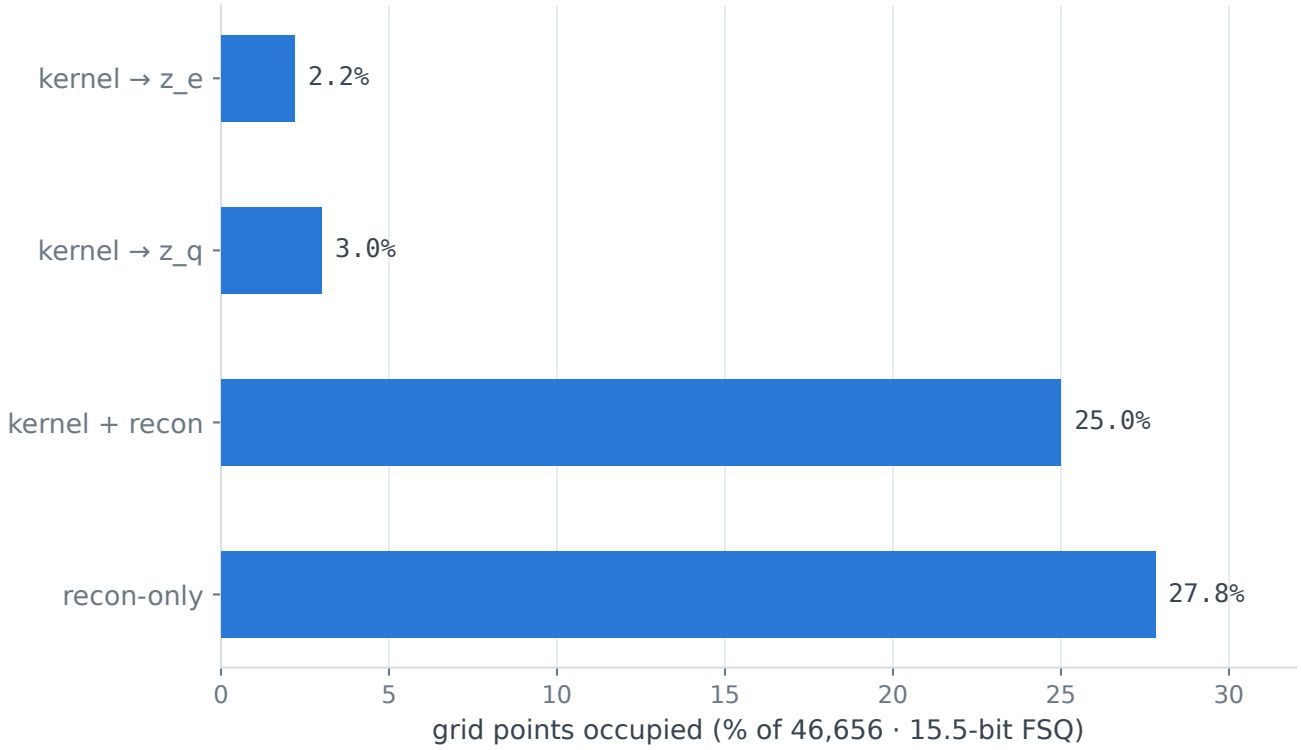


With no codebook, no EMA and no commitment loss, the kernel still degrades the FSQ token relative to plain reconstruction at 10, 15.5, 18 and 24 bits; both trained variants remain below the untrained grid.

Conclusion: the C4/C5 negative was not caused by codebook dynamics. Together, C4–C6 close the criterion-shaped discrete program: the conflict is between the kernel geometry and quantization itself.

AOJ · 20 epochs · kernel→z_q ablation: 0.092; kernel + recon w=1: 0.571 (15.5 b)

6c — mechanism: kernel-induced code degeneracy

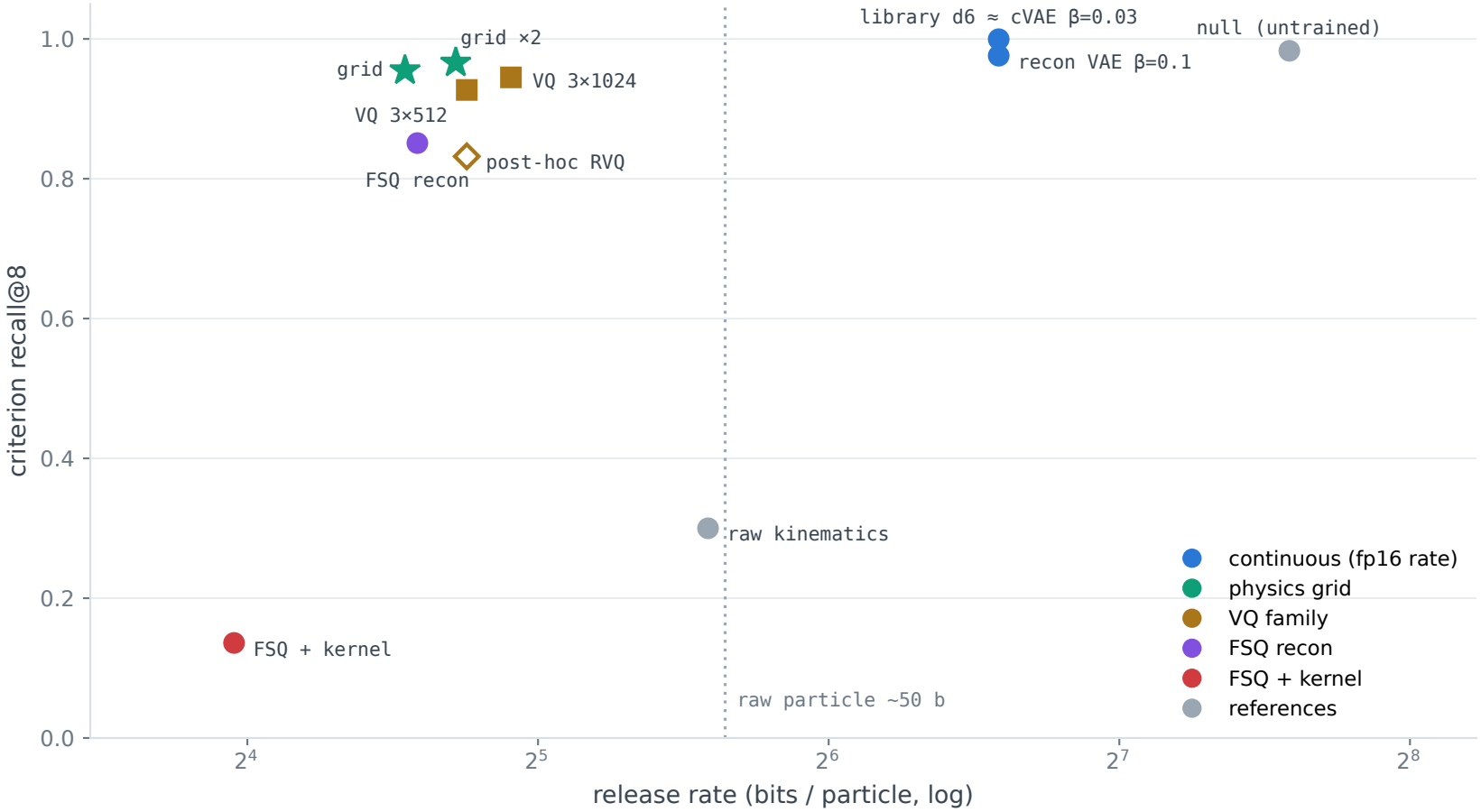


The kernel's fixed point drives criterion partners toward coincidence in the latent; any quantizer maps coincidence to identical codes. Kernel arms occupy ~2 % of the grid vs ~28 % for reconstruction.

Retrieval cannot rank true partners among thousands of identical-code impostors, so recall collapses. The mechanism is quantizer-independent — one cause for the C4 (RVQ) and C6c (FSQ) negatives.

Implication: install the criterion on a continuous latent; a discrete tier should quantize the features (6b), not the criterion-shaped embedding.

All release candidates on one axis (honest rates)



Continuous tokens at their fp16 release size (slide 18 convention); the 13.3 KL-bits of the β=0.03 arm is a diagnostic. Full map with every C1–6 arm: benchmarks/figures/library_rate.png.

Revisions to previously recorded conclusions

STATUS	CONCLUSION	EVIDENCE
revised	'reconstruction buys species, not the criterion' (C5)	β=1 rate artifact; 0.976 at β=0.1 (3 seeds)
relabelled	the '~1.4-bit' contrastive VAE (C5)	KL ≠ release rate for deterministic μ; quote fp16
closed	criterion-shaped discrete tokens (C4→C6c)	kernel hurts every quantizer at every rate tested
confirmed	criterion/information complementarity (C2b→C6a)	survives rate-matching: 0.9984 recall, $R^2 \leq 0.82$
new baseline	untrained physics grid (C6b)	0.966 @ 26.3 b — the bar for any discrete proposal

All pre-registered outcomes and full tables: benchmarks/README.md §Campaign 6; design and audit: docs/research/campaign6_alternate_schemes.md.

Conclusions and outlook

Conclusions

- The released-token decision reduces to a stated trade: criterion-organized and compact → the contrastive VAE / dim-6 library (recall 0.998, 96 fp16 b); a discrete tier → the untrained double grid (0.966 @ 26.3 b, R^2 0.949); maximal fidelity → plain VQ at 30 b (R^2 0.970).

- Installing the physics criterion requires a smooth continuous latent; quantizing a criterion-shaped embedding fails for a quantizer-independent reason (code degeneracy).

- Rate claims for continuous tokens must be operational (fp16 or a codec), not ELBO-derived.

Outlook

- Campaign 7 (runs today): masked-continuous FM probe over the release menu — no C3–C6 winner has a consumer-side number yet; this is the mission's own success metric.

- The open corner: REPA-style frozen-teacher distillation and the symbols ⊕ latent factorized release; plus d0/dz feature restoration and the protocol's anomaly-detection probe.

- North star (data-blocked): cross-experiment invariance — the context-free library and the grid tier are both constructed for it; a second-detector sample is the missing input.

References

VQ-VAE	van den Oord, Vinyals, Kavukcuoglu — Neural Discrete Representation Learning, arXiv:1711.00937
FSQ	Mentzer et al. — Finite Scalar Quantization: VQ-VAE Made Simple, arXiv:2309.15505
BSQ / LFQ	Zhao et al., arXiv:2406.07548 · Yu et al. (MAGVIT-v2), arXiv:2310.05737
VQ gradient fixes	rotation trick, arXiv:2410.06424 · SimVQ, arXiv:2411.02038
REPA / RAE	Yu et al., arXiv:2410.06940 · arXiv:2510.11690
continuous AR tokens	MAR, arXiv:2406.11838 · GIVT, arXiv:2312.02116 · JetFormer, arXiv:2411.19722
HEP tokenization	OmniJet-α, arXiv:2403.05618 · MPM, arXiv:2401.13537 · tokenization-free MPM, arXiv:2409.12589
SSL objectives	VICReg, arXiv:2105.04906 · JetCLR, arXiv:2108.04253 · HEP-JEPA, arXiv:2502.03933 · J-JEPA, arXiv:2412.05333
per-particle representations	EFN/PFN, Komiske, Metodiev, Thaler — arXiv:1810.05165
data	CMS Aspen Open Jets, RunG 2016 open data release
this work	benchmarks/README.md (campaign record) · docs/research/ (method of record, C6 design & audit)